

Esneider Ríos

Statistician – Universidad Nacional de Colombia, Medellín Campus

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Portfolio: esneider-rios-stats.vercel.app | GitHub: github.com/eariosb | LinkedIn: linkedin.com/in/esneider-rios

PROFESSIONAL PROFILE

Final-year Statistics student at Universidad Nacional de Colombia. Before returning to school, I spent over a decade running my own businesses in the retail and services sectors, directly responsible for budgets, inventory, and client and supplier relationships. Over the past two years I have combined that experience with training in Bayesian analysis, mixed models, time series and software development, building my own statistical applications — from model to deployment — in quantitative finance, retail pricing and LegalTech.

PROJECTS & APPLICATIONS IN PRODUCTION

Bayesian Trading System for SPY — BSTS + eGARCH(1,1)

Directional Bayesian model (Bayesian Structural Time Series, spike-and-slab prior) combined with an eGARCH model for volatility management. Out-of-sample validation over 2020–2026, with hyperparameter calibration performed exclusively on pre-2020 data and statistical significance tested via a Monte Carlo test ($p = 0.024$). Full technical documentation available in the portfolio.

La Tienda — Price Intelligence (Shiny)

Wholesale price-intelligence system for retail merchants in Antioquia. Conjugate Normal-Normal Bayesian updating, anomaly detection and 15-day forecasting, using data from Central Mayorista, DANE-SIPSA, Banco de la República, and the World Food Programme.

TimeSight (Shiny)

Time series analysis platform with walk-forward validation, ARIMA/ETS/TBATS models and Bayesian components, designed for users with no programming experience.

MLM Studio (Shiny)

Complete linear mixed models workflow (lme4): data loading, longitudinal profiles, estimation and comparison of nested models, with automatic natural-language interpretation.

"Legal Digital Transformation in Colombia" Course (LegalTech)

Original 20-hour course on jurimetrics, AI applied to law, and Colombian regulatory frameworks.

Code, technical articles and additional courses available through the links in the header.

KEY COMPETENCIES

- Statistical analysis and Bayesian modeling
- Data application development (R/Shiny, Python, Next.js)
- Financial, inventory and business management
- Communicating technical results to non-technical audiences

TECHNICAL TOOLS

Data analysis: R, Python, Bayesian analysis (BSTS, MCMC, spike-and-slab), mixed models (lme4), time series (eGARCH, ARIMA/ETS/TBATS), advanced Excel

Engineering & deployment: Docker, CI/CD (GitHub Actions), reproducible pipelines (targets, renv), APIs (Plumber), monitoring (Prometheus)

Web development: HTML, CSS, Tailwind, JavaScript, React, Next.js, Node.js, NestJS, Shiny

Databases: SQL (design, queries and modeling), MongoDB

Office: Microsoft Excel, PowerPoint and Word

EDUCATION

Universidad Nacional de Colombia

Statistics Undergraduate — Medellín Campus

In progress — final semesters

Training in statistical modeling, Bayesian analysis, information processing and interpretation, trend and behavior analysis, and insights oriented toward decision-making, alerts and forecasting in business and process-control settings.

Academlo — Full Stack Web Development

2023 — 2024

End-to-end web application design and development: back-end with Node.js, Express and NestJS; front-end with HTML, CSS, Tailwind, ReactJS and Next.js.

PROFESSIONAL EXPERIENCE

Independent Merchant

Apr 2020 — Apr 2024

Own business · Food & beverage retail

Full management of a consumer-goods trading business. Responsible for financial planning, budgeting, inventory control, margin analysis and sales forecasting.

Business Management Consultant

Feb 2007 — Apr 2012

Shawarma Restaurant

Consulting on cost management, inventory optimization, and the selection and implementation of administrative software solutions.

Manager and Owner

Sep 2016 — Dec 2019

D'Mente Inoxidable · Metal fabrication workshop

Operational, commercial and financial management of a metal-fabrication workshop. Purchasing, cost structures, per-project budgeting and profitability control.

Owner and Manager

Apr 2012 — Sep 2018

DistriKuwait · Trading company

Led the full commercial operation: staff selection and management, purchasing, inventory, budgets and business-control software tools.

Transportation & Logistics Operations — United States

Sep 2024 — Dec 2025

VTFirstClass Ride · RNGTransport (client: Advance Auto Parts) · Delivery platforms

Executive transportation, inter-store merchandise logistics, and order-cycle management on delivery platforms.

ADDITIONAL COURSES

Fundamentals of Senior Management (2025) · Ethics at Work and Data Analysis (2025) · Assertive Communication (2025) · Emotional Intelligence (2025) · SQL Databases (2023)

Software Engineer Fundamentals (2025) · Mentoring and Coaching for Leaders (2025) · Influence Techniques (2025) · Developing Empathy (2025)

LANGUAGES

Spanish: Native | **English:** Upper-Intermediate · B2